MSc Thesis Direction

Context-Aware and Agentic Anomaly Detection for SIEM Environments

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PART 1 — COMPLETED WORK

Ranking-Based Anomaly Detection for SOC Alert Triage on Wazuh SIEM

Problem

At a Managed Detection & Response provider, SOC analysts drown in alerts. Most daily Wazuh events are medium-severity, with a long tail of low-signal rules, and analyst capacity is fixed. Rather than treat detection as a binary classification over the whole corpus, I reframed it as a triage budget: if an analyst only reads the top 1% of each day, the task is to maximise true positives inside that slice. Metrics followed the same logic — precision@p, recall@p, lift@p, and FPR@p, all computed per day.

Approach

- **Dataset:** 277,499 Wazuh alerts from a production CSOC index (10 Jul – 12 Aug 2025), 5 agents, 2,359 unique source IPs.
- **Split:** T1 training window (214,538 rows) → T2 validation with curated MITRE ATT&CK-style synthetic injections at 1–3%/day (36,022 rows) → T3 prospective hold-out (26,880 rows). Feature transformers fit on T1 only and frozen — no leakage.
- **Features:** temporal (hour, off-hours, weekend, bucketed periodicity), severity (per-host z-score of rule.level), rarity (time-since-last rule-on-host, hashed agent–rule–hour signatures), plus light interactions.
- **Models compared:** k-NN (distance-based scoring), Isolation Forest, LOF.
- **Explainability:** per-alert reason codes listing top contributing features, neighbour-median comparator, and context flags (e.g. off-hours × severity) to speed analyst hand-off.

Results (p = 1% daily operating point, T3 hold-out)

Model	Config	Precision@1%	Recall@1%	Lift@1%	FPR@1%
k-NN	k=35	0.6801	0.2340	23.34×	0.0033
Isolation Forest	600 trees, 0.5 samples	0.4032	0.1388	13.84×	0.0061
LOF	k=20	0.3010	0.1036	10.33×	0.0072

k-NN beat Isolation Forest by ~1.69× and LOF by ~2.26× on precision and lift. On T3, the 1%-per-day slice gives 253 true positives against 119 false positives (TP:FP ≈ 2.13:1). Per-day precision and lift stay in a tight interquartile range — stable without re-tuning.

Deployment & Impact

The ML layer runs as a FastAPI + Uvicorn scoring service with frozen feature transformers for train/inference parity. It plugs into the Telkom MDR stack — Wazuh SIEM, Suricata NIDS, DFIR-IRIS case management, and YARA/ClamAV/VirusTotal threat-intel — so Top-K rankings feed analyst queues and n8n SOAR automation. At the 1%-per-day operating point, analysts catch 23× more true positives than random sampling, turning a flat backlog into a ranked, explainable shortlist.

Context-Aware and Agentic Extension

Motivation

My BSc thesis showed that ranking-based anomaly detection works in production SIEM — but the ranking layer currently operates without behavioural context (user roles, peer-group baselines, temporal norms), and the output is a static list that analysts must interpret alone. Modern SIEM threats — especially insider attacks — go undetected for months because rule-based defences lack this behavioural dimension [1][2]. Meanwhile, over 80% of analysts hesitate to act on anomaly alerts without explanations [3]. I propose extending the ranking layer into a **context-aware anomaly detection system with an agentic support model** that provides explainable insights and semi-autonomous alert triage, integrating with SOC workflows to close the loop from detection to response.

Objectives

- **Context-Aware Anomaly Detection:** Incorporate contextual information (user roles, temporal patterns, peer-group behaviour) into the ML anomaly detection to catch complex threats such as insider attacks that evade static rule sets.
- **Agentic Analyst Support:** Design an intelligent agent to assist SOC analysts via semi-autonomous alert triage and investigation. The agent dynamically prioritises and correlates alerts based on evolving context [4], provides human-readable explanations for anomalies, and recommends response actions.
- **SOC Workflow Integration:** Ensure seamless integration with existing SOC tools (SIEM, SOAR, ticketing) to automate routine tasks, accelerate incident response, and reduce MTTR.

Proposed Methods

The approach combines unsupervised machine learning with rule-based context modelling: k-NN models (validated in Part 1) learn baseline behaviour patterns to flag deviations, and each alert is enriched with contextual data — peer-group baselines, historical norms, and threat intelligence. An LLM-based agent performs alert triage: using retrieval-augmented generation, it gathers relevant log data and past cases to explain an alert's anomaly [3]. In a human-in-the-loop setup, the agent automatically dismisses low-risk anomalies and escalates high-risk alerts with explanations. A prototype integrated with a SOAR platform will trigger containment actions (e.g. disable compromised accounts) and create incident tickets. Evaluation targets: detection accuracy, false-positive rate, and operational metrics (alert handling time, MTTR) using CERT insider-threat logs and simulated SOC data.

Background and Prior Work

Context-aware anomaly detection has shown improved effectiveness — adding contextual information to user-behaviour sequences enhanced detection of insider threats such as privilege abuse [2]. The RAPID system dynamically prioritises alerts, reducing investigation time by up to 99% [4], while LogRESP-Agent uses an AI assistant to generate context-rich explanations for alerts [3]. I propose unifying these advances into one cohesive system integrating anomaly detection, context enrichment, and an explainable triage agent aligned with SOC processes.

Timeline and Milestones (6 months)

- **Months 1–2:** Literature review; collect log data; define context features; set up SIEM/SOAR environment.
- **Months 3–4:** Develop and tune anomaly detection models (unsupervised ML + contextual rules); implement context enrichment modules; prototype the explanation agent.
- **Months 5–6:** Refine the agent's triage logic with feedback; integrate with SOAR and ticketing workflows; evaluate on an insider-threat scenario to quantify detection and response improvements.

Directions This Thesis Naturally Opens

The proposed system sits at an intersection that admits several natural extensions, each of which could deepen the thesis or seed follow-on research:

- **Attack-graph integration** — feeding ranked anomaly outputs into probabilistic attack-graph models so alert triage conditions path-likelihood estimates rather than being consumed as isolated events.
- **Active learning for ground-truth closure** — converting analyst triage decisions into a growing real-incident label set, enabling threshold recalibration, drift detection, and robustness against evasive distributions. This is the route from the current ranking layer toward real-time situational awareness.
- **Agentic security testing** — the semi-autonomous triage agent's architecture generalises to AI-assisted penetration testing and red-team support, where an agent proposes test vectors and interprets results with human oversight.
- **LLM-agent repair pipelines** — explainable reason codes from the ranking layer become structured inputs for agents that propose, and under guardrails execute, remediation on the top-ranked anomalies.
- **Supply-chain and SBOM anomaly detection** — the same ranking framing and rarity features port to software-composition and build-time events, where base rates and budget constraints follow the same shape.

References

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